

UPDATED RECENT-LITERATURE BENCHMARK REPORT

3-Stack Silicon Nanosheet GAAFET - Sentaurus TCAD

Re-analysis against very recent 2024-2026 papers, with emphasis on 2025-2026 results

Device	3 stacked Si nanosheets, NMOS GAAFET
Geometry	Lg = 10 nm; sheet thickness = 4 nm
Gate stack	SiO2/HfO2 dual dielectric; metal-gate work-function tuning
Isolation	Bottom dielectric isolation (BDI) over p-Si substrate
Reported FOMs	SS 65.0 mV/dec; DIBL 10.32 mV/V; Vth,sat 0.070 V
Current metrics	Ion 1.438 mA/um; Ioff 0.216 pA/um; Ion/Ioff about 6.6e9

Prepared from the supplied benchmark report plus a fresh literature search completed 6 September 2026

1. Executive conclusion

Bottom line

Your electrical results are not broadly "96% accurate" in the strict calibration sense, but several key numbers are physically credible. SS = 65 mV/dec is strongly supported by recent experimental and TCAD literature. DIBL = 10.32 mV/V is exceptionally good but still falls inside the best recent experimental/simulation range (about 7-14 mV/V for optimized structures). Ion = 1.438 mA/um is plausible for an optimized or idealized 10 nm GAAFET and is within about 5% of a recent 10 nm high-k strained-GAAFET simulation. The main credibility problem is the extremely low Ioff, which drives an unrealistically large Ion/Ioff ratio unless interface traps, GIDL/BTBT, contact parasitics, and process variability are explicitly modeled.

The supplied report correctly identifies the strength of the electrostatics, but it overstates the global accuracy. On page 1, the report itself shows a 63.14% DIBL deviation from its old benchmark and a 68.75% threshold-voltage deviation, while later text describes SS and DIBL as matching within 2.3%. Those two statements cannot both be true. The updated report therefore replaces a single "accuracy percentage" with metric-by-metric validation.

What changed after checking recent papers

- SS remains the strongest result: 65 mV/dec agrees almost exactly with recent high-quality GAAFET data.
- DIBL is better than many 10 nm TCAD references, but comparable to the best optimized SOI/SiGe experimental GAAFETs; it should be treated as an excellent result that still needs extraction verification, not as proof of overall model accuracy.
- Ion is credible in magnitude, but current normalization must be stated explicitly because literature uses both effective gated perimeter and device footprint width.
- Ioff is 3.1 decades below a recent fabricated Omega-SOI GAAFET and 4.1 decades below a 10 nm high-k strained-GAAFET simulation. This is the largest evidence of idealization in the present deck.
- Recent 2025-2026 work shows that BDI is valuable but not free: full BDI suppresses leakage, while it can interrupt stress transfer and reduce drive current. A stress-recovered/partial BDI study is therefore a strong next research direction.
- The original explanation that self-heating produces an upward Id-Vds tilt should be corrected. Recent electrothermal studies report that self-heating raises lattice temperature, degrades mobility, and reduces on-state/saturation current; an upward high-Vds slope is more directly associated with output conductance/channel-length modulation and drain electrostatics.

2. Baseline extracted from your supplied report

The current benchmark report describes a 3-stack silicon nanosheet NMOS with 4 nm sheet thickness, 10 nm physical gate length, a dual SiO₂/HfO₂ gate dielectric, BDI, and a p-type silicon substrate. It reports SS = 65.0 mV/dec, DIBL = 10.32 mV/V, V_{th,sat} = 0.070 V, Ion = 1.438 mA/um, Ioff = 0.216 pA/um, and log₁₀(Ion/Ioff) = 9.82.

Metric	Your value	Initial interpretation
SS	65.0 mV/dec	Near-ideal electrostatic swing

Metric	Your value	Initial interpretation
DIBL	10.32 mV/V	Very strong drain-field immunity
V _{th,sat}	0.070 V	Very low threshold; dominated by work function/extraction
Ion	1.438 mA/um	Strong drive current
I _{off}	0.216 pA/um	Extremely low room-temperature leakage
Ion/I _{off}	~6.6 x 10 ⁹	Mostly driven by the very low I _{off}

Important consistency issue in the old report

The old page-1 table gives DIBL = 10.32 mV/V versus 28 mV/V, which is a -63.14% relative difference. Therefore the later claim that "SS and DIBL match within 2.3%" is mathematically incorrect. Only the SS comparison was within about 2.3% of that particular old benchmark.

3. Very recent papers selected for the updated benchmark

Priority was given to 2025-2026 papers and to devices that overlap with your geometry, material stack, layer count, BDI strategy, or 10 nm gate length. Papers with different channel materials or pFET polarity are used only for electrostatic context, not as strict one-to-one calibration targets.

1. Bopparathi, Jukanti, Konda & Vadakattu, 2026 - IEEE GSEACT

Three stacked silicon nanosheets, TiN/HfO₂ gate, HfO₂ thickness 0.78 nm. Reported V_{th} = 0.40 V, SS = 63 mV/dec, DIBL = 46 mV/V, Ion = 1e-4 A and I_{off} = 1e-12 A. This is one of the closest 2026 TCAD architectures to your project.

2. Amudalapalli et al., 2026 - Journal of Physics: Conference Series

Compares two-channel and three-channel nanosheet devices at the 10 nm node using TCAD. Uses Si channel, SiO₂ and HfO₂ dielectrics and evaluates V_{th}, SS, DIBL, Ion, I_{off} and transfer/output curves. The indexed abstract confirms extremely high structural relevance, but the accessible indexing used for this report did not expose enough numeric FOMs for a fair numerical deviation table.

3. Sun et al., 2025 - Microelectronics Journal, Omega-SOI GAAFET

Fabricated four-layer SOI nanosheet GAAFET with leakage suppression. Reported SS = 65 mV/dec, DIBL = 7 mV/V, I_{off} = 2.8e-4 uA/um (280 pA/um) and Ion/I_{off} = 3e6. This is an important experimental reference for the best achievable electrostatics.

4. Kim et al., 2025 - Communications Engineering (Nature portfolio)

Large-scale full-quantum simulation of realistic stacked nanosheet GAAFETs. Shows that current gain saturates around 3-4 stacks, simplified structures can overestimate Ion, and quantum access resistance at the S/D-to-nanosheet constriction can strongly reduce current. The work benchmarks against recent hardware and notes the highest TSMC reference exceeds 1 mA/um.

5. Ding et al., 2025 - IEEE 3M-NANO

Sentaurus TCAD study of source/drain bottom dielectric isolation in Si nanosheet GAAFETs. BDI suppresses

sub-fin leakage and improves gate-capacitance behavior. This directly supports the physical motivation for your BDI layer.

6. Improved uniformity and performance via full BDI, 2026 - Microelectronics Journal

Compares stacked Si nanosheet GAAFETs on bulk and SOI with full bottom dielectric isolation. Full BDI reduces I_{off} by about one order of magnitude and improves V_t uniformity. Contact/landing-pad optimization can raise nFET I_{on} by about 121%, emphasizing the role of parasitic source/drain resistance.

7. Stress-recovered BDI, 2026 - Microelectronics Journal

Shows a modern tradeoff that is directly relevant to your project: full BDI cuts bottom leakage but interrupts S/D stress transfer. A stress-recovered BDI architecture improves NMOS I_{on} by 18% relative to Full-BDI at $iso-I_{off} = 0.1$ nA while retaining channel isolation.

8. Jia et al., 2026 - IEEE Transactions on Electron Devices

Fabricated four-level Si_{0.7}Ge_{0.3} nanosheet GAAFET with improved release/passivation. Reports $SS = 71$ mV/dec, $DIBL = 14$ mV/V and $I_{on}/I_{off} = 4.7e6$. Different channel material and pFET operation, but an excellent experimental reference for realistic interface quality and electrostatic control.

9. Perumal et al., 2024 - Micromachines

10 nm GAAFET with a tri-layer strained Si/SiGe channel and stacked SiO₂/HfO₂ gate dielectric with EOT about 0.7 nm. Reports V_{th} about 0.298 V, $I_{on} = 1.516$ mA/um, $I_{off} = 2.66$ nA/um, $SS = 72.42$ mV/dec and $DIBL = 56.24$ mV/V. It remains one of the most useful recent quantitative 10 nm references for your I_{on} .

4. Quantitative metric-by-metric benchmark

The percentages below are numerical proximity indicators only. They are not universal "accuracy" values because the papers use different channel materials, isolation schemes, bias definitions, contact assumptions, and current-normalization conventions.

Reference	SS (mV/dec)	DIBL (mV/V)	V_{th} (V)	I_{on} (mA/um)	I_{off} (pA/um)	I_{on}/I_{off}	Comparability
Your TCAD	65.0	10.32	0.070	1.438	0.216	6.6e9	Baseline
Bopparathi 2026	63	46	0.40	not normalized	not normalized	not normalized	Very high structural
Omega-SOI 2025	65	7	-	-	280	3e6	Experimental electrostatics
Perumal 2024	72.42	56.24	0.298	1.516	2660	5.71e5	High for 10 nm current
Jia SiGe 2026	71	14	-	-	-	4.7e6	Electrostatic context
Comparison	SS deviation		DIBL deviation		V_{th} deviation		I_{on} deviation
vs Bopparathi 2026	+3.2%		-77.6%		-82.5%		n/a
vs Omega-SOI 2025	+0.0%		+47.4%		n/a		n/a
vs Perumal 2024	-10.2%		-81.7%		-76.5%		-5.1%
vs Jia 2026	-8.5%		-26.3%		n/a		n/a

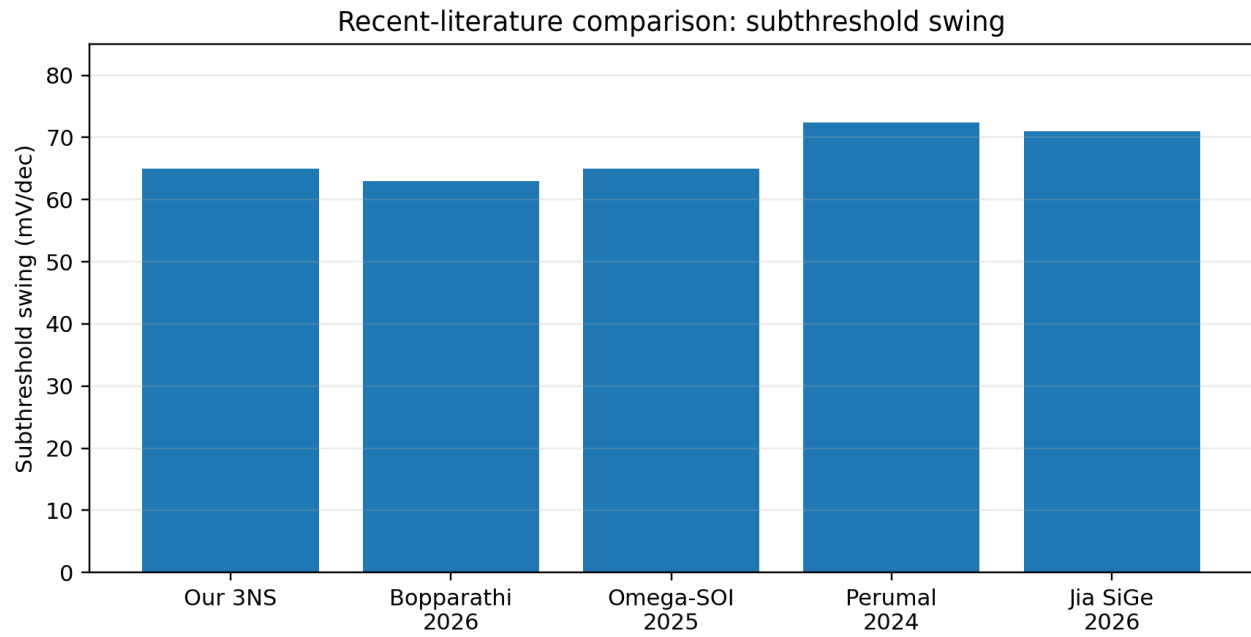


Figure 1. SS comparison. Your 65 mV/dec result sits inside the strongest recent GAAFET range.

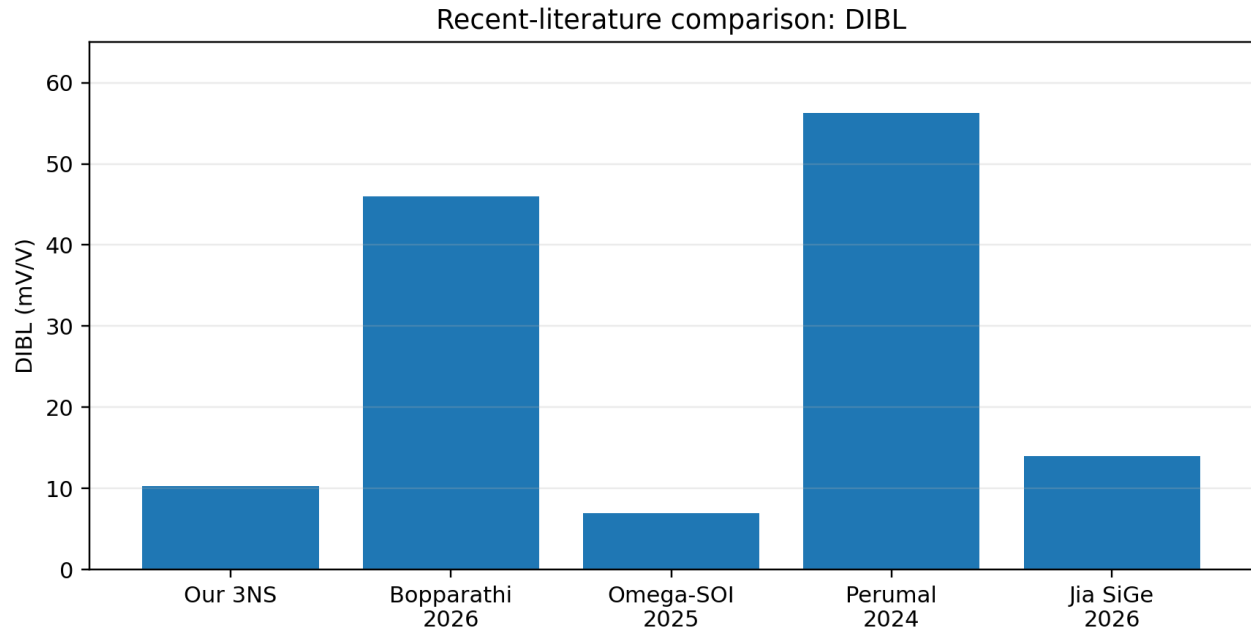


Figure 2. DIBL comparison. Your 10.32 mV/V is far better than many 10 nm TCAD designs, but close to optimized experimental 7-14 mV/V references.

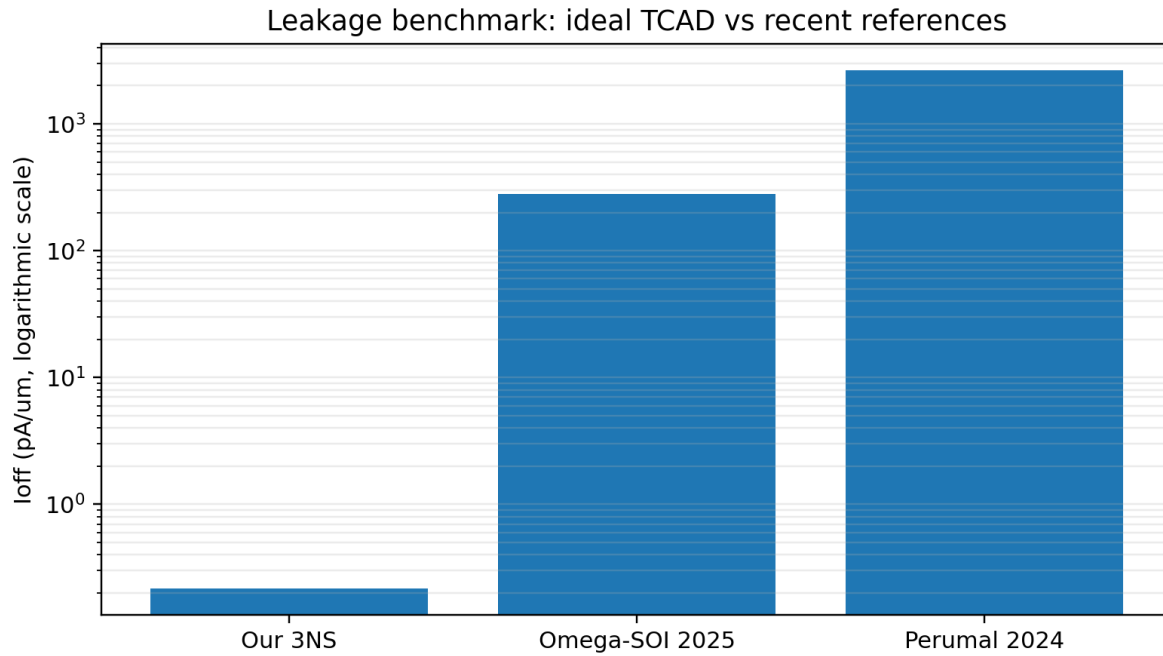


Figure 3. Leakage comparison on a log scale. The very low I_{off} is the clearest sign that the present TCAD deck is idealized.

5. Detailed physical interpretation

5.1 Subthreshold swing: strong and credible

$SS = 65$ mV/dec is the most defensible result in the current report. It is identical to the 2025 fabricated Omega-SOI result and only about 3.2% above the 63 mV/dec value in the 2026 three-stack Si/HfO₂ TCAD paper. It is also better than the 71-72.4 mV/dec values reported in recent SiGe/strained-channel references. For a 4 nm sheet with full GAA wrap, strong gate control is expected.

However, SS should be re-extracted from the log I_d - V_g curve using a clearly documented current window. If the current curve contains a discontinuity or solver transition near threshold, an apparently excellent SS can be artificially produced by the extraction interval.

5.2 DIBL: exceptional, but not impossible

DIBL = 10.32 mV/V is much lower than the 46-56 mV/V values found in recent 10 nm TCAD references, yet it is in the same order as optimized experimental structures: 7 mV/V for the fabricated Omega-SOI device and 14 mV/V for the 2026 SiGe stacked GAAFET. Therefore the value should not be rejected simply because it is very low. Instead, it should be validated by extraction consistency and geometry/bias checks.

- Use the same constant-current V_{th} extraction method at low and high V_{DS} .
- Report the exact two V_{DS} values used for DIBL.
- Check whether the BDI or substrate contact is unintentionally over-isolating the electrostatics.
- Repeat with fine mesh and a second quantum model setting to demonstrate numerical stability.

5.3 Threshold voltage: the present 0.070 V is not technology-calibrated

Your $V_{th,sat} = 0.070$ V is far below recent 10 nm/5 nm TCAD references around 0.298-0.40 V. This does not mean the simulation is wrong; V_{th} is intentionally tunable through metal work function, fixed charge, channel doping, EOT and extraction method. But a 70 mV threshold is too low to use as evidence of N2-class realism without a stated target and a documented work-function calibration.

Recommended wording

Call $V_{th} = 0.070$ V a "work-function-tuned simulation operating point," not a validated technology-node threshold. For publication, add a V_{th} calibration sweep versus gate work function and select a target V_{th} from a chosen benchmark.

5.4 On-current: surprisingly close to a recent 10 nm simulation

$I_{on} = 1.438$ mA/ μ m differs by only about -5.1% from the 1.516 mA/ μ m reported for the 10 nm strained Si/SiGe high-k GAAFET of Perumal et al. This makes the magnitude credible. The 2025 full-quantum stacked-GAAFET study also notes that the strongest recent hardware reference from TSMC exceeds 1 mA/ μ m, while many realistic devices fall below that because of source/drain access resistance and quantum-geometrical effects.

The important qualification is normalization. Some GAAFET papers normalize current to total gated perimeter (effective width), while others normalize to footprint width. A 1.4 mA/ μ m number can change substantially depending on that definition. The updated report should explicitly state the normalization equation and the raw drain current before normalization.

5.5 Off-current and I_{on}/I_{off} : currently too ideal

$I_{off} = 0.216$ pA/ μ m is about 1,296 times lower than the 280 pA/ μ m fabricated Omega-SOI reference (3.11 decades) and about 12,315 times lower than the 2.66 nA/ μ m 10 nm high-k strained-GAAFET simulation (4.09 decades). This enormous difference is not explained by electrostatics alone.

The likely causes are the ideal assumptions already acknowledged in your report: zero/very low interface-trap density, omission of GIDL and BTBT, ideal S/D junctions, no trap-assisted tunneling, no random variability, and possibly overly ideal contacts. Because the I_{on} value is already realistic in magnitude, the huge I_{on}/I_{off} ratio is mainly a consequence of suppressed leakage rather than a uniquely superior transport design.

5.6 Output characteristics and self-heating: correction required

The old report attributes perfectly flat saturation to an isothermal 300 K model and says experimental upward tilt is due to self-heating. The first half is reasonable as a simplification, but the second half should be corrected. Recent electrothermal GAAFET studies report that self-heating increases lattice temperature, increases scattering, reduces carrier mobility and decreases I_{on} /saturation current. A positive slope in saturation is more directly related to finite output conductance, channel-length modulation, DIBL and high-field electrostatics. Including thermodynamic self-heating will generally reduce the current relative to the isothermal result, not create a simple upward tilt.

6. What the newest BDI papers imply for your structure

BDI is now one of the most relevant parts of your project because multiple 2025-2026 studies directly analyze it. These recent papers strengthen your structure concept but also reveal a missing tradeoff in the current report.

1. Leakage benefit: recent full-BDI work reports about one order of magnitude lower I_{off} and improved V_t uniformity compared with bulk GAAFETs.
2. Parasitic benefit: the 2025 IEEE BDI study reports suppression of sub-fin leakage and improvements in gate-capacitance behavior.

3. Stress penalty: the newest 2026 stress-recovered BDI work shows that Full-BDI can interrupt source/drain stress transfer into the channel, reducing mobility and Ion.
4. Recovery route: retaining a controlled Si path beneath S/D while preserving dielectric isolation beneath the channel can recover current; the 2026 SR-BDI study reports +18% NMOS Ion versus Full-BDI at iso-Ioff.

Direct research opportunity for your project

Instead of treating BDI as a fixed block, sweep BDI thickness and dielectric constant, then compare Full-BDI, partial/source-drain BDI, and stress-recovered BDI. Extract Ion, Ioff, SS, DIBL, Cgg, substrate leakage and lattice temperature. This is more publishable than a single fixed-BDI device because recent literature has established a real leakage-performance tradeoff.

7. Implications of the 2025 full-quantum stacked-GAAFET paper

Kim et al. (2025) is especially important for interpreting your next simulation stage. Their large-scale NEGF analysis shows that simplified nanosheet models can overestimate current because the transition from bulk S/D to a confined nanosheet creates quantum access resistance. They also show that current scaling gives diminishing benefit after roughly 3-4 stacked sheets, which directly supports the engineering relevance of your three-stack architecture.

- Do not assume that a lower classical source/drain resistance always produces higher Ion; abrupt constriction geometry can increase quantum carrier reflection.
- Source/drain extension geometry and constriction doping should be part of optimization, not just the gate/channel.
- A drift-diffusion + density-gradient deck is appropriate for a baseline, but publication claims at 10 nm should acknowledge quasi-ballistic/quantum-access effects that are not fully captured.
- The present Ion being close to 1.5 mA/um may be an ideal upper-range result. Adding realistic S/D access and contact effects could reduce it while preserving the strong SS/DIBL.

8. Updated validation scorecard

Metric/claim	Updated assessment	Confidence	Required action
SS = 65 mV/dec	Excellent and literature-supported	High	Re-extract with stated current window
DIBL = 10.32 mV/V	Excellent; comparable to best optimized structures	Medium-high	Verify extraction, mesh and VDS pair
Vth = 0.070 V	Physically possible but not node-calibrated	Medium	Gate-work-function calibration sweep
Ion = 1.438 mA/um	Plausible; close to 10 nm high-k TCAD	Medium-high	State width normalization and add contact resistance
Ioff = 0.216 pA/um	Too ideal for fabricated room-temperature comparison	Low as hardware prediction	Add Dit, BTBT/GIDL and junction realism
Ion/Ioff about 6.6e9	Mathematically correct but dominated by ideal leakage	Low as hardware benchmark	Recalculate after leakage calibration
Flat Id-Vd saturation	Acceptable isothermal baseline	Medium	Add thermodynamic/SHE run

Metric/claim	Updated assessment	Confidence	Required action
>96% overall accuracy	Not supported by the available comparison	Low	Replace with metric-specific validation

9. Minimum simulation additions before calling the model publication-calibrated

- Current normalization audit: report raw I_d and normalized I_d using a clearly defined width (footprint width and/or total gated perimeter).
- Threshold calibration: sweep gate work function and select a target V_{th} supported by one chosen recent benchmark; keep the rest of the geometry fixed.
- Leakage calibration: add interface trap density, BTBT/GIDL where appropriate, and at least one non-zero D_{it} sensitivity sweep.
- Contact/access resistance: include realistic contact resistance or source/drain series resistance and compare with the ideal-contact baseline.
- Self-heating experiment: compare isothermal drift-diffusion with thermodynamic transport and report ΔT plus peak lattice temperature.
- Transport experiment: compare drift-diffusion with hydrodynamic/energy-transport if stable in your Sentaurus version; treat it as a sensitivity study rather than a direct substitute for full NEGF.
- Mesh independence: coarse/medium/fine meshes must keep V_{th} , SS, DIBL and I_{on} within a small tolerance (for example 2-3%).
- BDI study: sweep BDI thickness/material and at least one partial/stress-recovered geometry; quantify leakage versus drive-current tradeoff.
- Statistical realism: if time permits, add process variation or interface-charge sensitivity so the model does not represent only a perfect nominal device.

10. Strongest research direction after the recent-paper update

The most defensible next paper is not simply "a 3-stack GAAFET with good I_d - V_g / I_d - V_d ." Recent literature already contains many such demonstrations. A stronger contribution would combine your existing stable TCAD structure with a modern isolation/transport question that the newest papers show is still active.

Recommended paper theme

Physics-guided multi-objective optimization of BDI in a 3-stack 10 nm silicon nanosheet GAAFET: jointly optimize BDI thickness/permittivity, source-drain extension, gate work function and nanosheet geometry for I_{on} , I_{off} , SS, DIBL and thermal behavior; then use the simulation database for a physics-guided surrogate model. The recent 2025-2026 BDI and full-quantum papers give a strong technical reason for this direction.

11. Final assessment

Your project has a solid electrical baseline. The updated literature does not invalidate the device; it changes which claims are safe. The strongest claims are excellent SS, excellent DIBL, and a credible high I_{on} . The weakest claims are the extremely low leakage, the resulting I_{on}/I_{off} , the very low threshold used as a technology benchmark, and the old single-number "96% accuracy" statement.

A scientifically stronger conclusion is: "The present idealized Sentaurus model reproduces the electrostatic and drive-current range of recent stacked-GAAFET literature, while leakage and threshold-voltage calibration remain intentionally idealized/tunable and require defect, tunneling, contact, thermal and process-nonideality models before direct hardware prediction."

12. Recent references used in this update

14. N. Bopparathi, S. Jukanti, S. N. Konda, and S. K. Vadakattu, "Design and Analysis of GAA Nanosheet FET Technology Node at 5nm," 2026 IEEE Global Symposium on Emerging and Communication Technologies (GSEACT), 2026. DOI: 10.1109/GSEACT68539.2026.11620143.
15. V. V. S. Amudalapalli et al., "Comparative Analysis of Two-Channel and Three-Channel Nanosheet FinFETs at 10 nm Node Using TCAD simulation," Journal of Physics: Conference Series, vol. 3196, 012039, 2026. DOI: 10.1088/1742-6596/3196/1/012039.
16. L. Sun et al., "A novel Omega-SOI Gate-All-Around FET with doping free load-Si and two-step wet etching achieving superior leakage suppression and short-channel effects immunity," Microelectronics Journal, vol. 156, 106534, 2025. DOI: 10.1016/j.mejo.2024.106534.
17. K. Y. Kim et al., "Quantum transport through a constriction in nanosheet gate-all-around transistors," Communications Engineering, vol. 4, article 92, 2025.
18. H. Ding, Z. Guo, Y. Yu, and H. Tian, "Source/Drain Bottom Dielectric Isolation in Si Nanosheet GAAFETs: A Simplified Scheme for Performance Improvement," 2025 IEEE International Conference on Manipulation, Manufacturing and Measurement on the Nanoscale (3M-NANO), pp. 549-552, 2025. DOI: 10.1109/3M-NANO65639.2025.11261006.
19. "Improved uniformity and performance of GAAFETs via full bottom dielectric isolation and quasi-self-aligned landing pads," Microelectronics Journal, vol. 173, 107174, 2026. DOI: 10.1016/j.mejo.2026.107174.
20. "A stress-recovered bottom dielectric isolation scheme for GAA nanosheet transistors with self-aligned backside contact integration," Microelectronics Journal, 107296, 2026. DOI: 10.1016/j.mejo.2026.107296.
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23. Y. Li et al., "Comprehensive electrothermal characterization of shrunk nanosheets in gate-all-around field-effect transistors," Micro and Nanostructures, vol. 206, 208213, 2025. DOI: 10.1016/j.micrna.2025.208213.
24. S. Rathore et al., "Systematic performance benchmarking of nanosheet and FinFET: An intrinsic self-heating perspective," Microelectronics Reliability, vol. 165, 115588, 2025. DOI: 10.1016/j.microrel.2025.115588.

Note on comparability: The papers above do not all use identical bias conditions, current normalization, gate length, channel material, contact model, substrate, EOT or extraction methodology. They are therefore used to define a recent credible performance envelope, not as a single direct calibration target.